Spatial Prisoner’s Dilemma

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How do cooperative strategies evolve in a CA simulation of prisoner’s dilemma?

Overview

- Introduce prisoner's dilemma
- Introduce Nowak and May (1992)
- Nowak and May's results
- Additional results

What is prisoner's dilemma?

- Thought experiment developed in 1950 by mathematicians Merrill Flood and Melvin Dresher of RAND Corporation
- Two members of a criminal gang are arrested, separated, and given two choices: stay silent (cooperate) or testify against the other (defect)
    - Testifying reduces jail time
- **Equilibrium:** the criminals will defect

Prisoner's Dilemma Payoff Matrix

1 \ 2	C	D
C	-1, -1	-3, 0
D	0, -3	-2, -2

Incentive to Defect

1 \ 2	C
C	-1, -1
D	0, -3

Incentive to Defect

1 \ 2	C
C	-1, -1
D	0, -3

Incentive to Defect

1 \ 2	D
C	-3, 0
D	-2, -2

Incentive to Defect

1 \ 2	D
C	-3, 0
D	-2, -2

Incentive to Defect

1 \ 2	C	D
C	-1, -1	-3, 0
D	0, -3	-2, -2

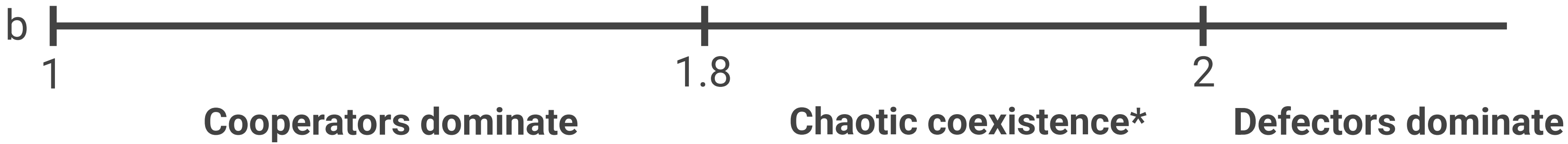
Modified Prisoner's Dilemma (Nowak and May)

1 \ 2	C	D
C	1, 1	0, b
D	b, 0	0, 0

“Evolutionary games and spatial chaos” (1992)

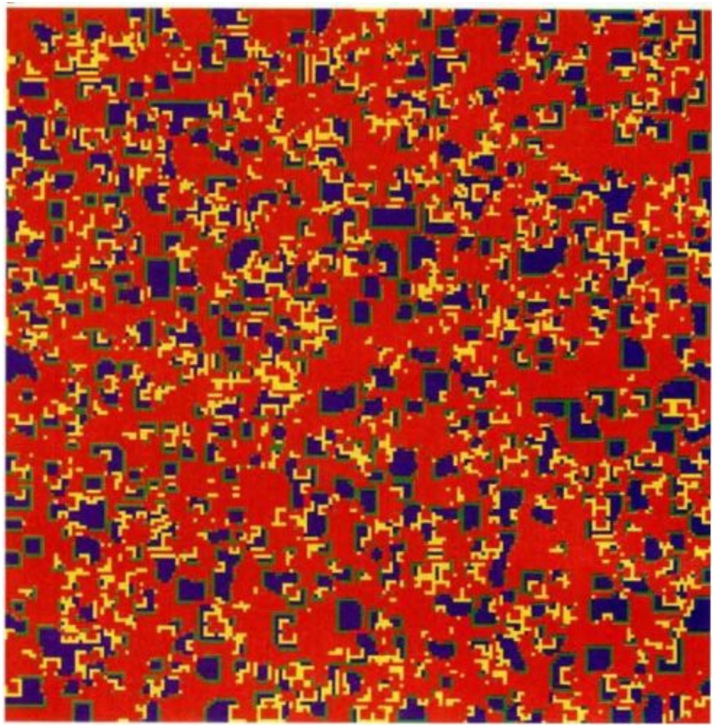
- Nowak and May: CA modified prisoner’s dilemma on 200 x 200 grid
- Each cell initialized to either cooperator (90%) or defector (10%)
- Each time step:
  - Agents play modified PD with each cell in Moore neighborhood
  - Agents accumulate rewards
  - Agents copy strategy of highest-payoff neighboring cell
- Vary parameter b and study evolution of cooperation

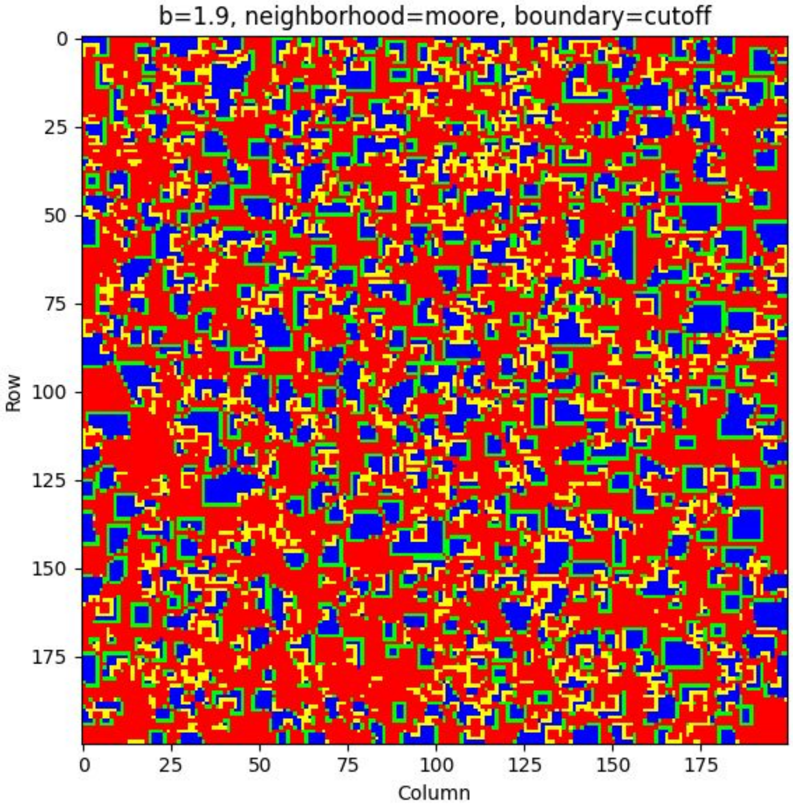
Nowak and May's Results



* Nowak and May found that at $1.8 < b < 2$, the proportion of cooperators settles to 0.32 independently of initial conditions

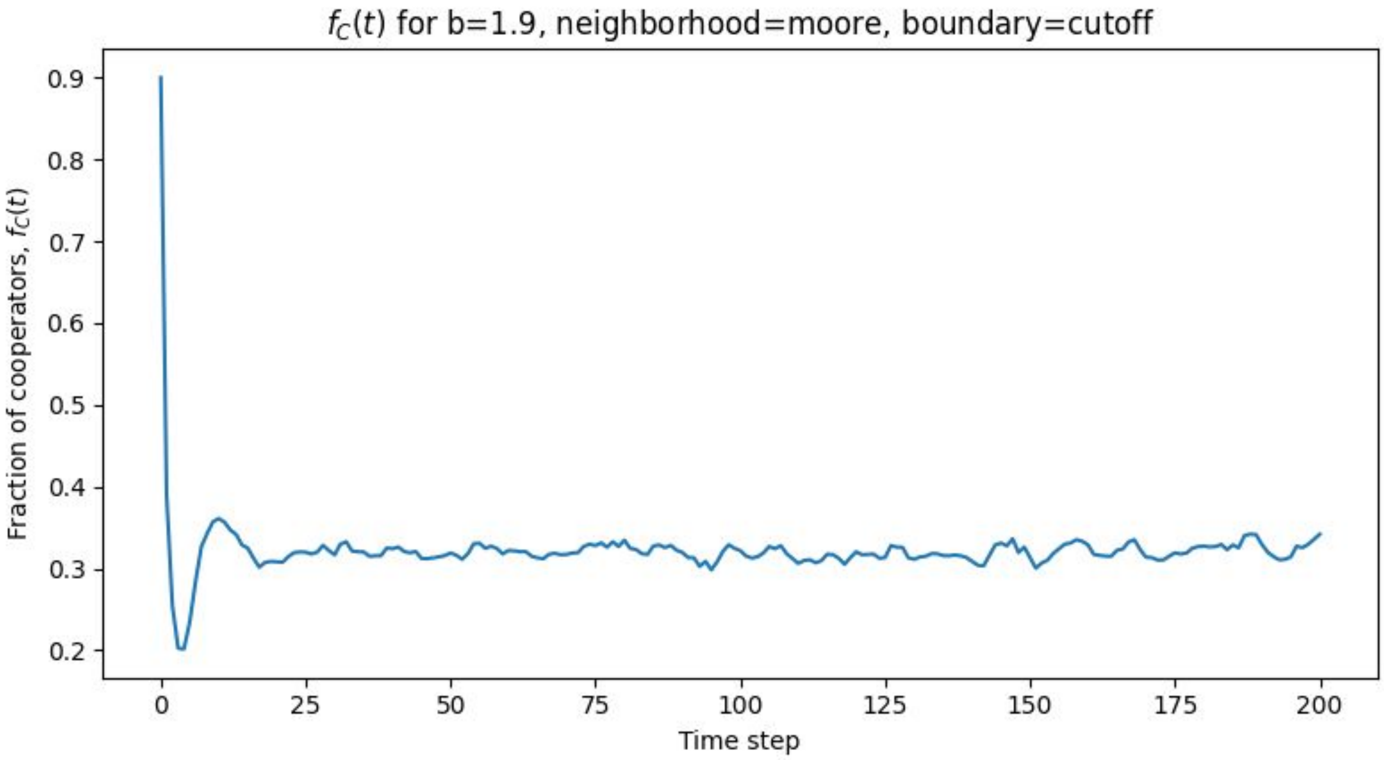
Results (b = 1.9)





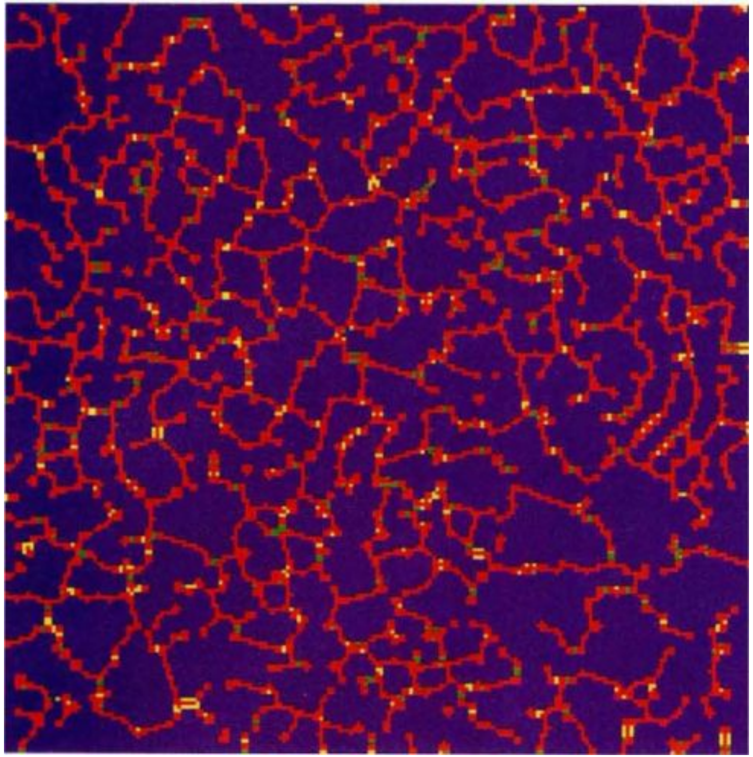
b=1.9, neighborhood=moore, boundary=cutoff
0
25
50
75
100
125
150
175
Row
0
25
50
75
100
125
150
175
Column

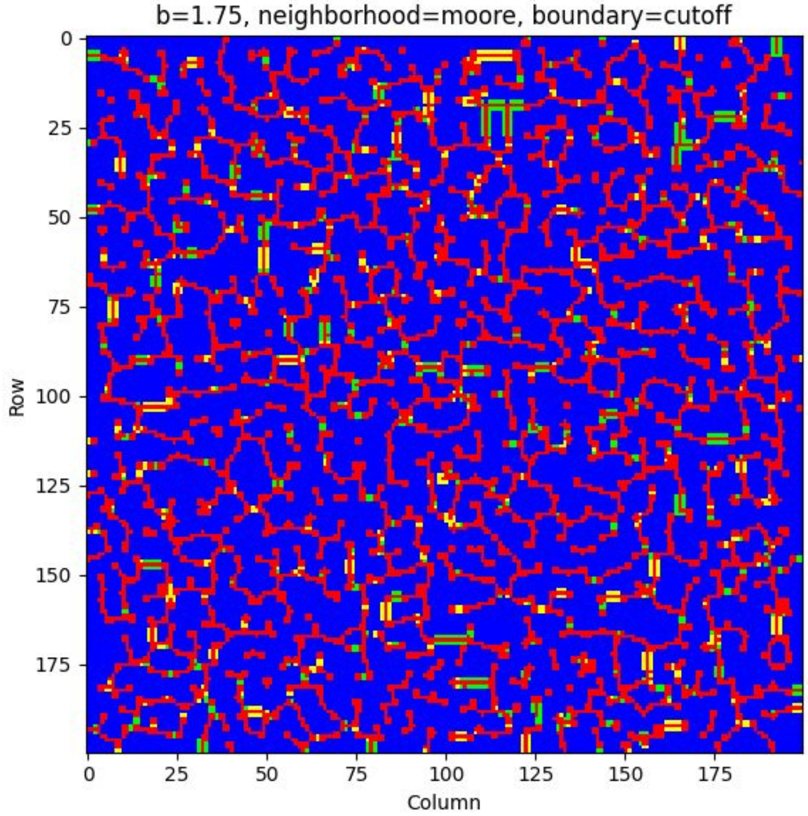
Results (b = 1.9)



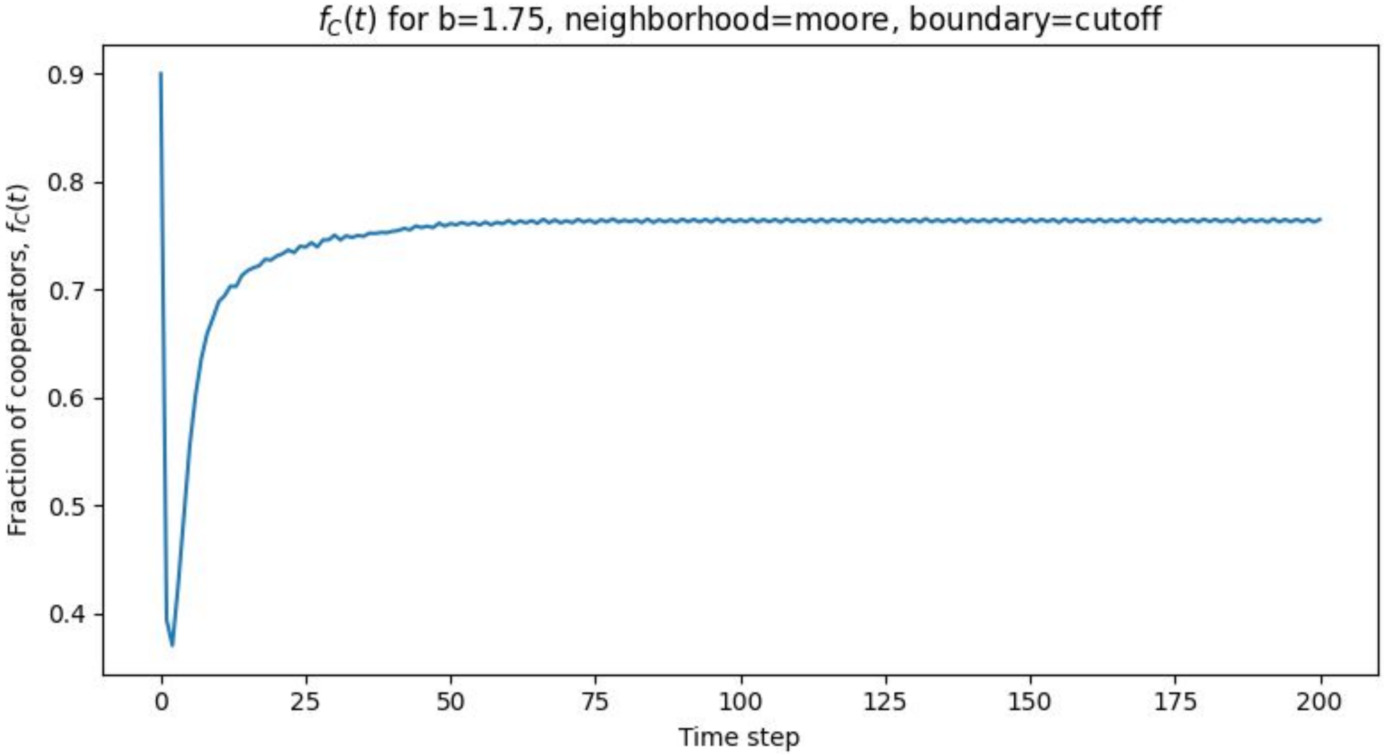
$f_C(t)$ for b=1.9, neighborhood=moore, boundary=cutoff
Fraction of cooperators, $f_C(t)$
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
25
50
75
100
125
150
175
200
Time step

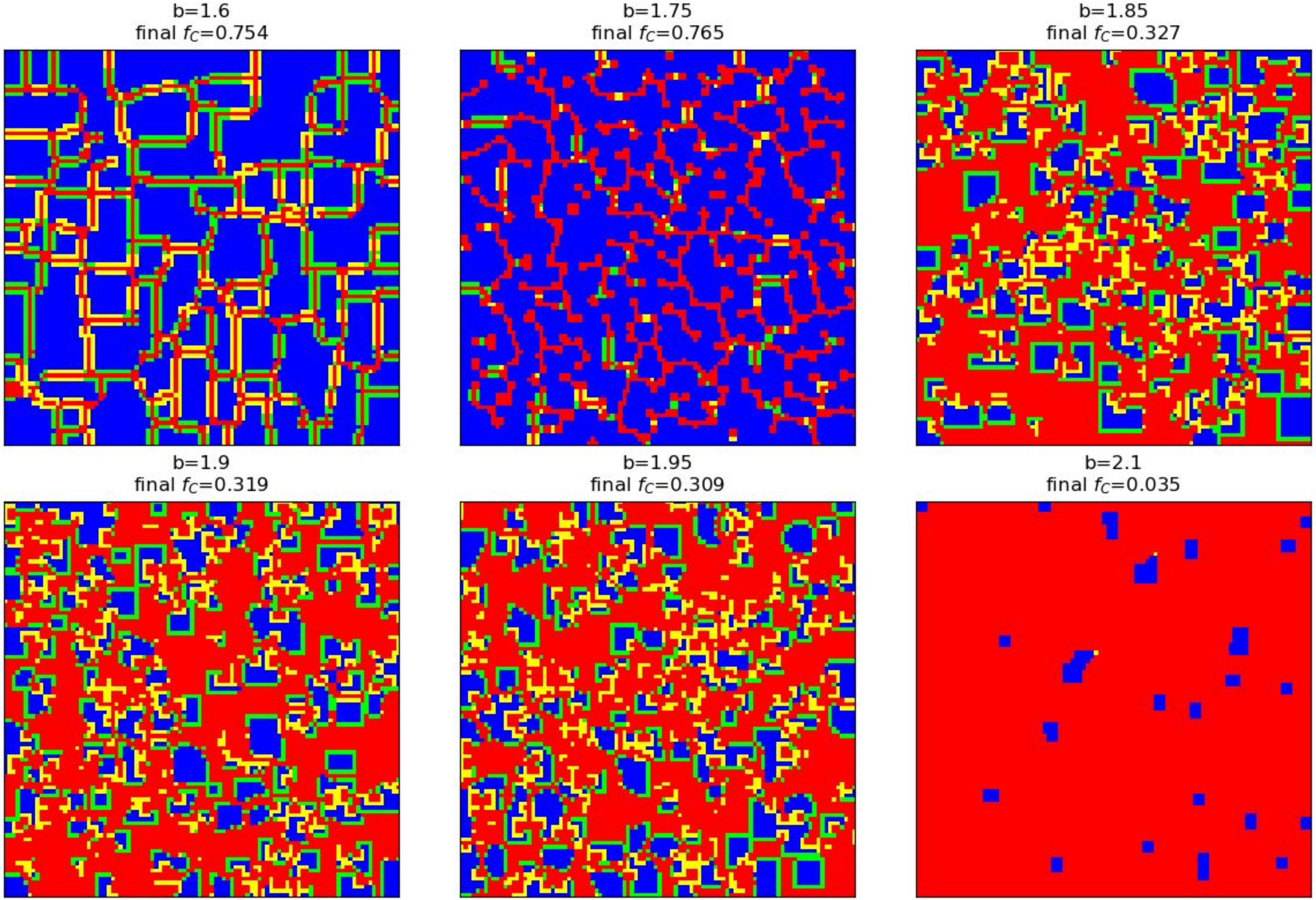
Results (b = 1.75)

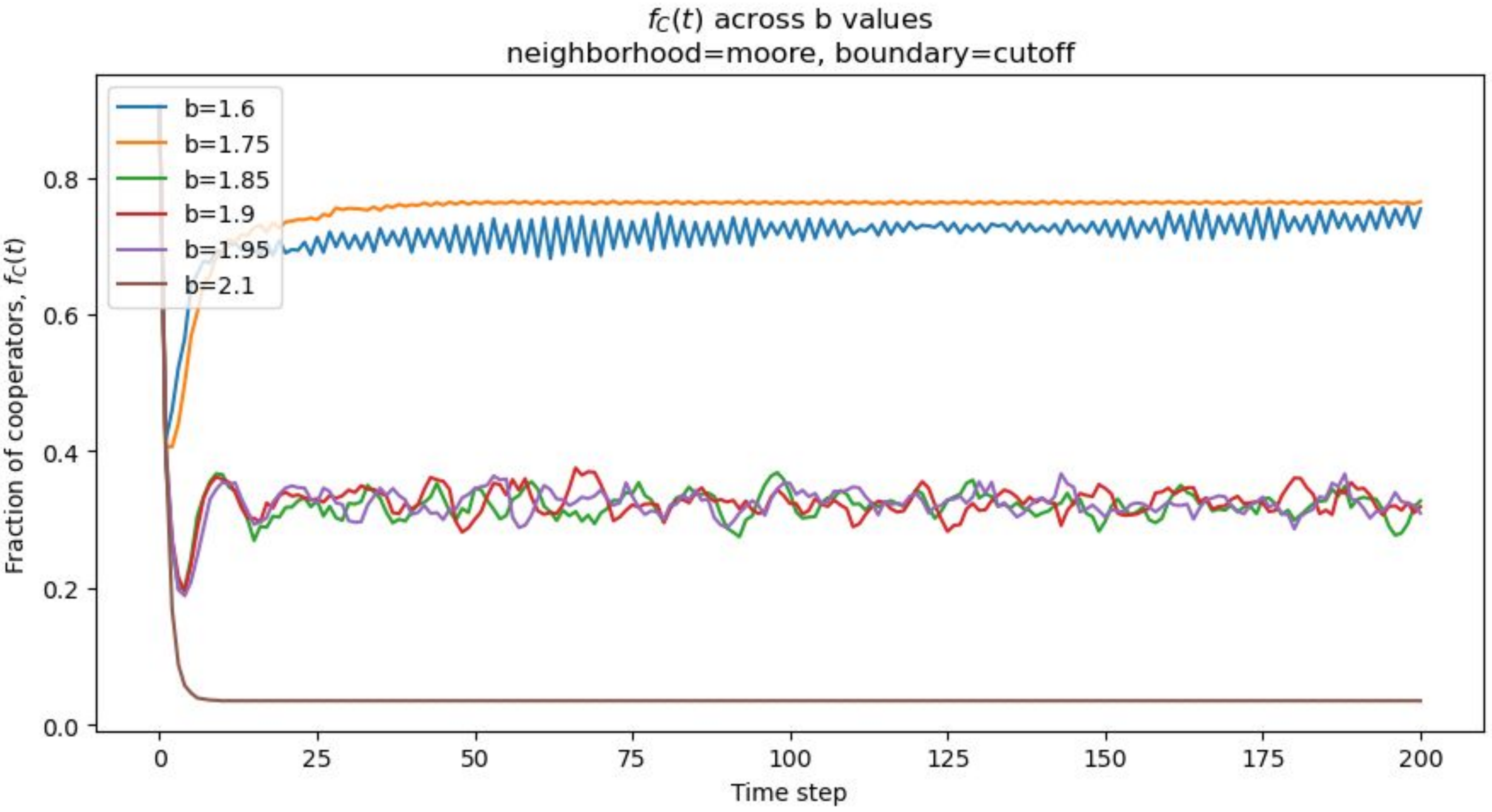




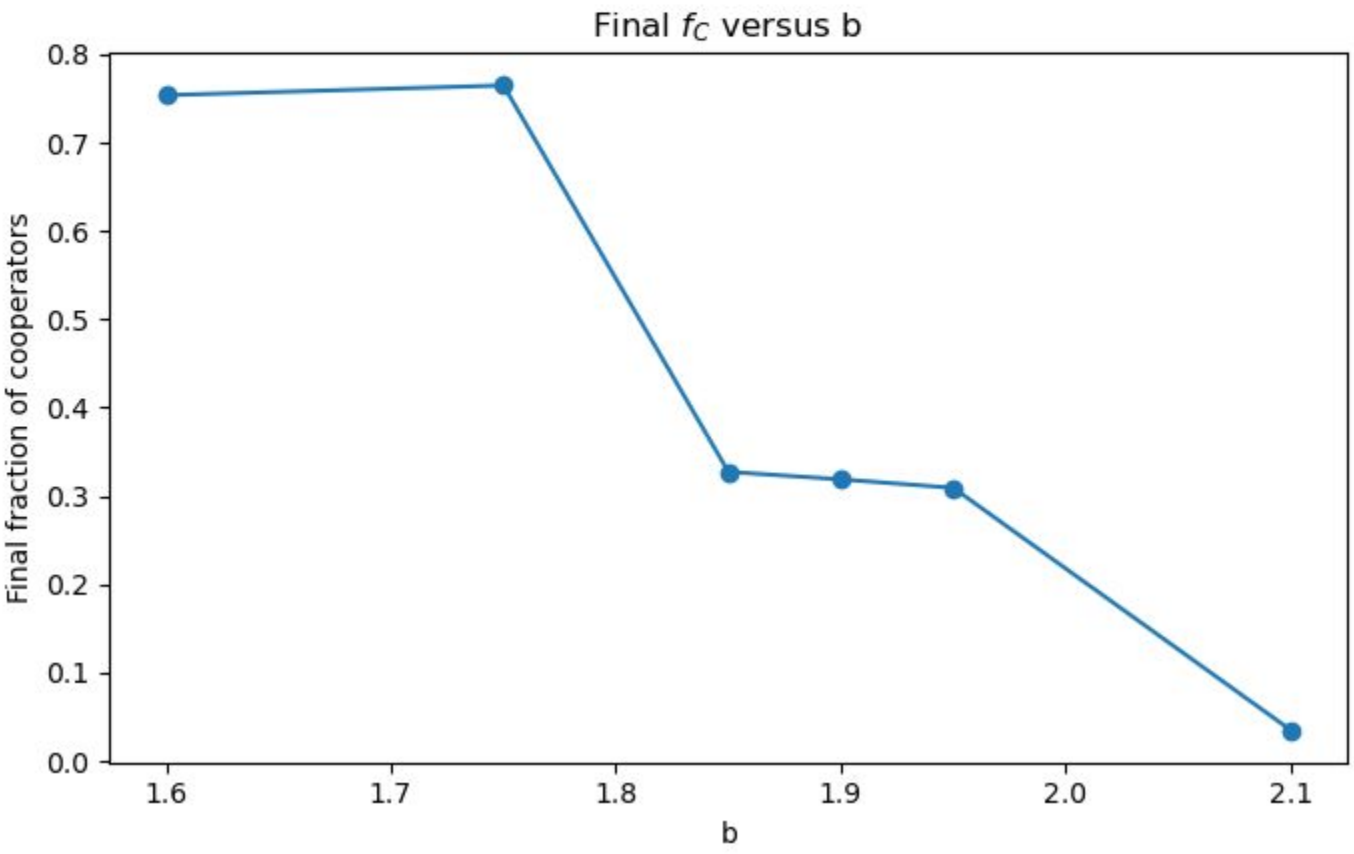
Results (b = 1.75)




b=1.6
final f_C=0.754
b=1.75
final f_C=0.765
b=1.85
final f_C=0.327
b=1.9
final f_C=0.319
b=1.95
final f_C=0.309
b=2.1
final f_C=0.035



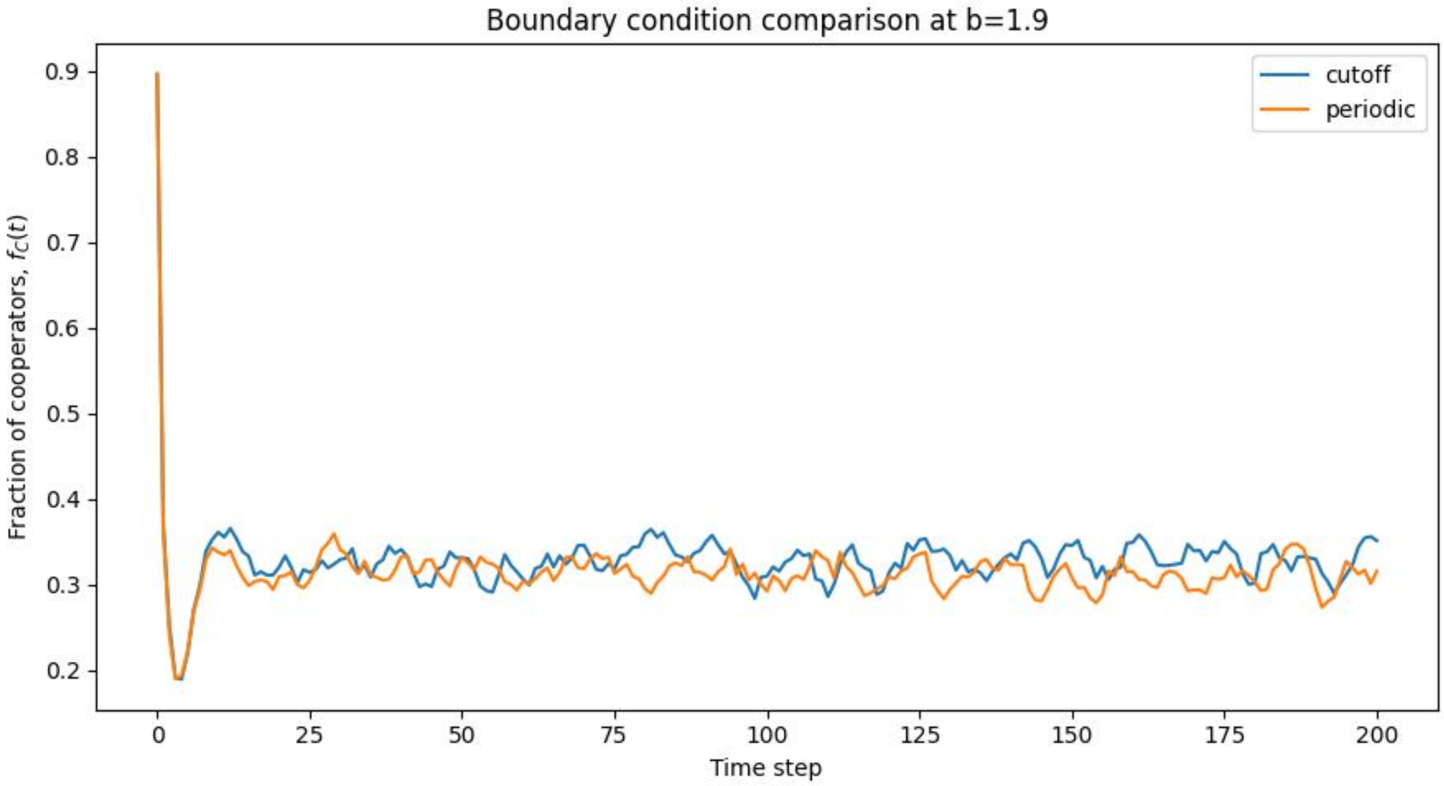
$f_C(t)$ across b values
neighborhood=moore, boundary=cutoff
b=1.6
b=1.75
b=1.85
b=1.9
b=1.95
b=2.1
Fraction of cooperators, $f_C(t)$
0.0
0.2
0.4
0.6
0.8
0
25
50
75
100
125
150
175
200
Time step


Final f_C versus b
Final fraction of cooperators
b
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
1.6
1.7
1.8
1.9
2.0
2.1

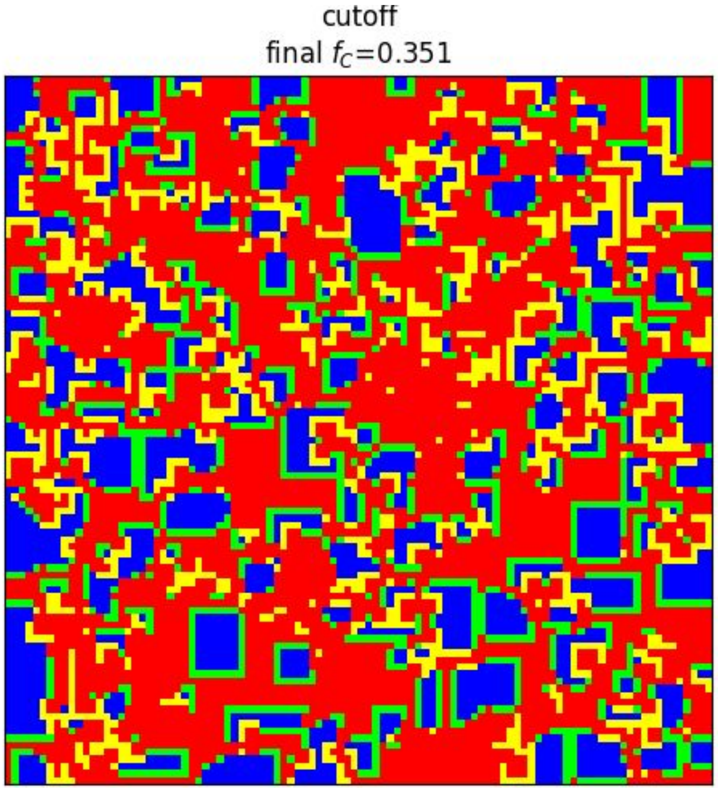
My Experiments

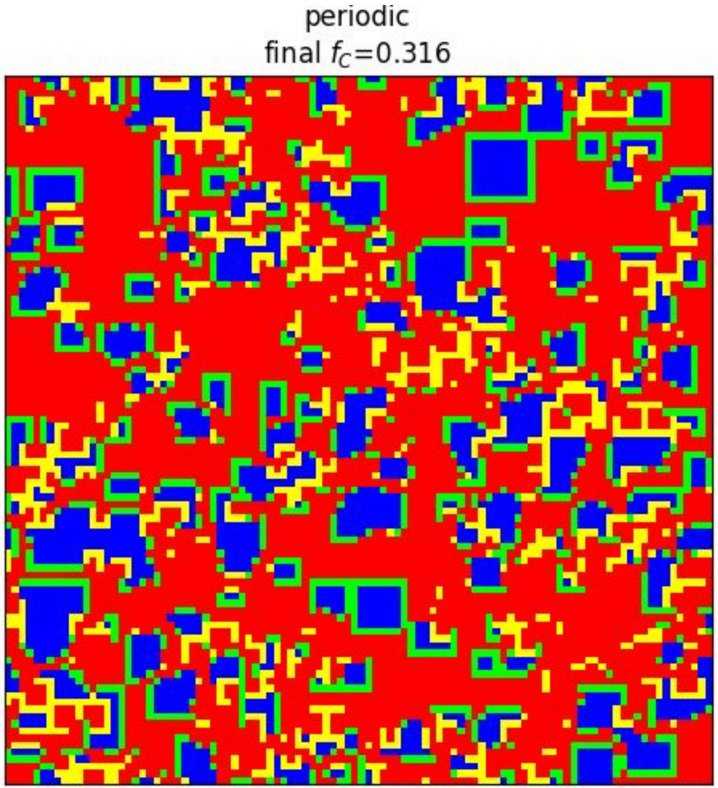
1. Do boundary conditions affect the equilibrium?
2. Do neighborhood variants affect the equilibrium?

Boundary Conditions (b = 1.9)

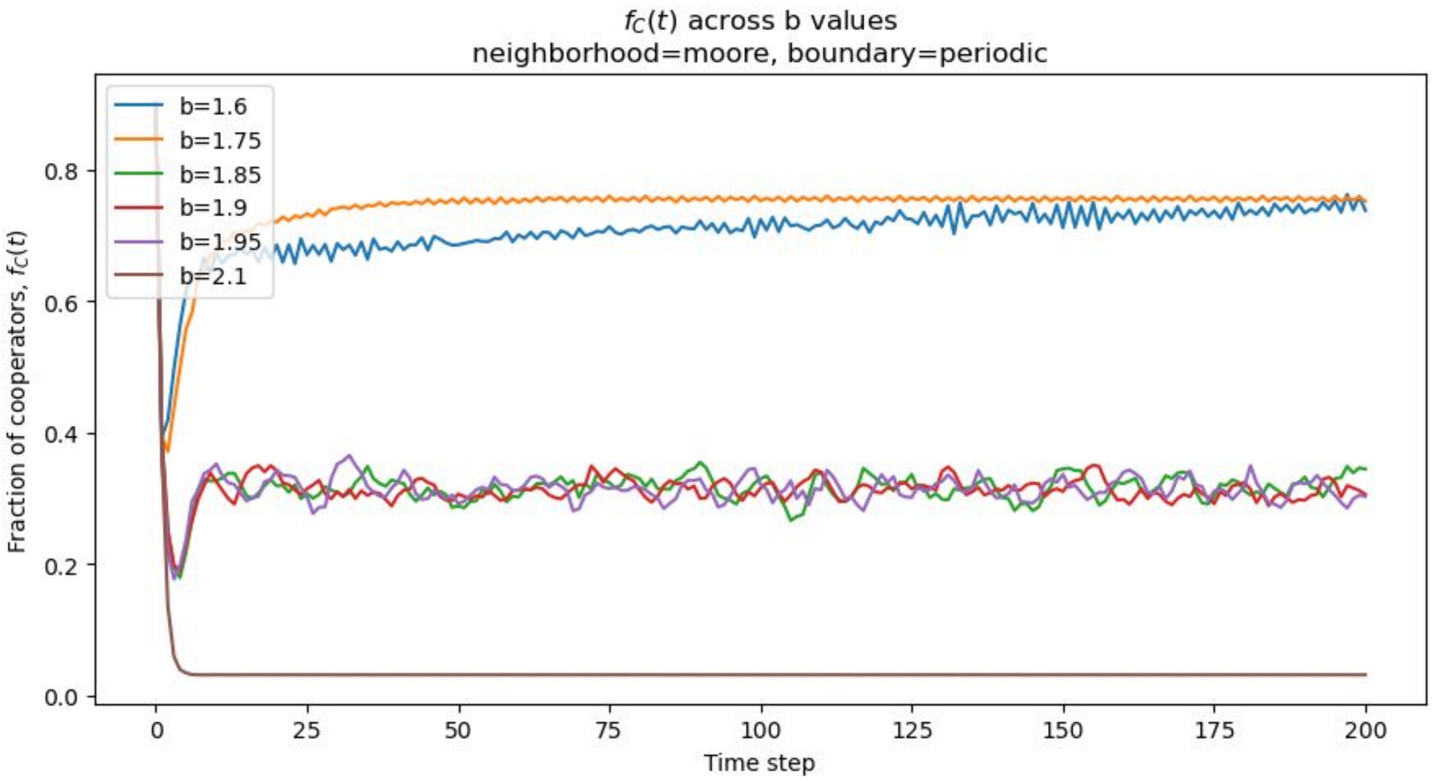


Boundary Conditions (b = 1.9)





Boundary Conditions (Vary b)

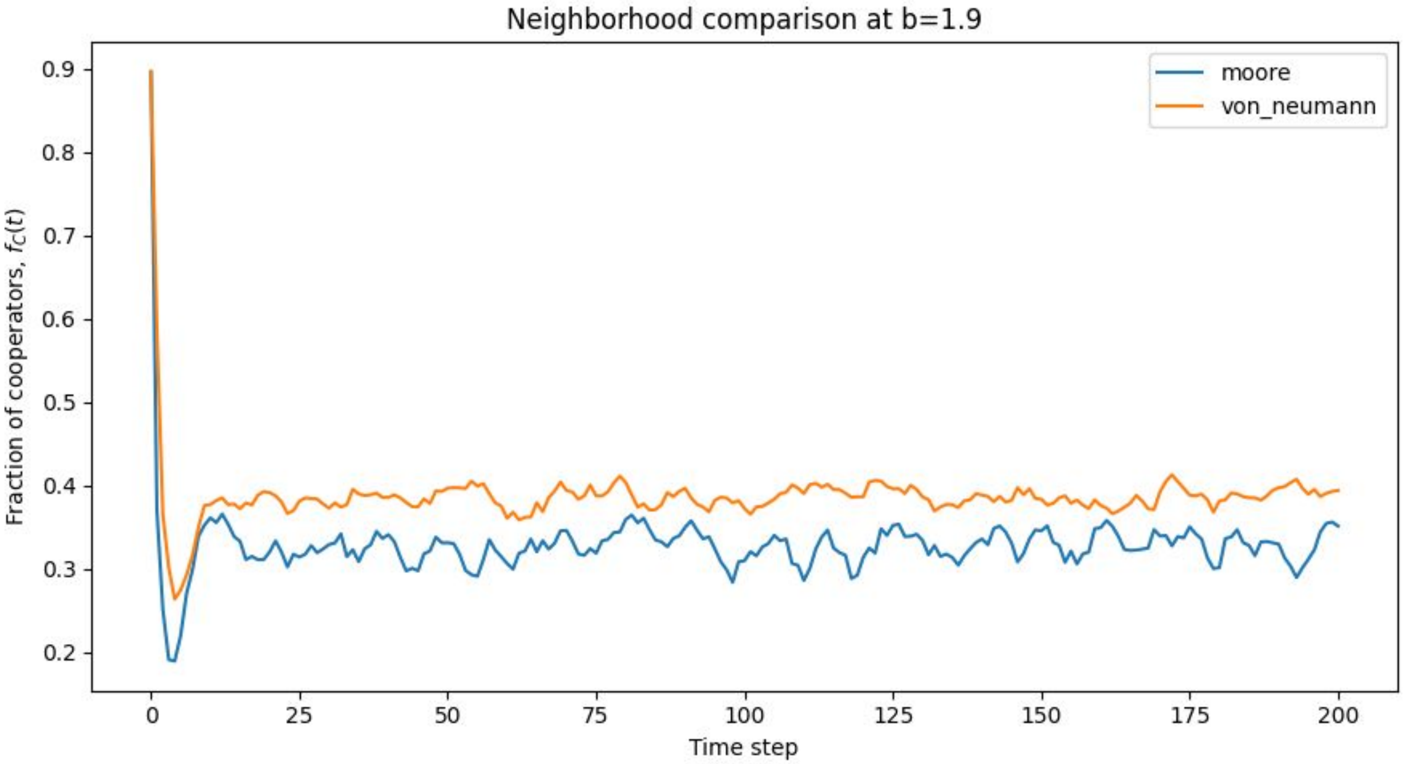


Boundary Conditions

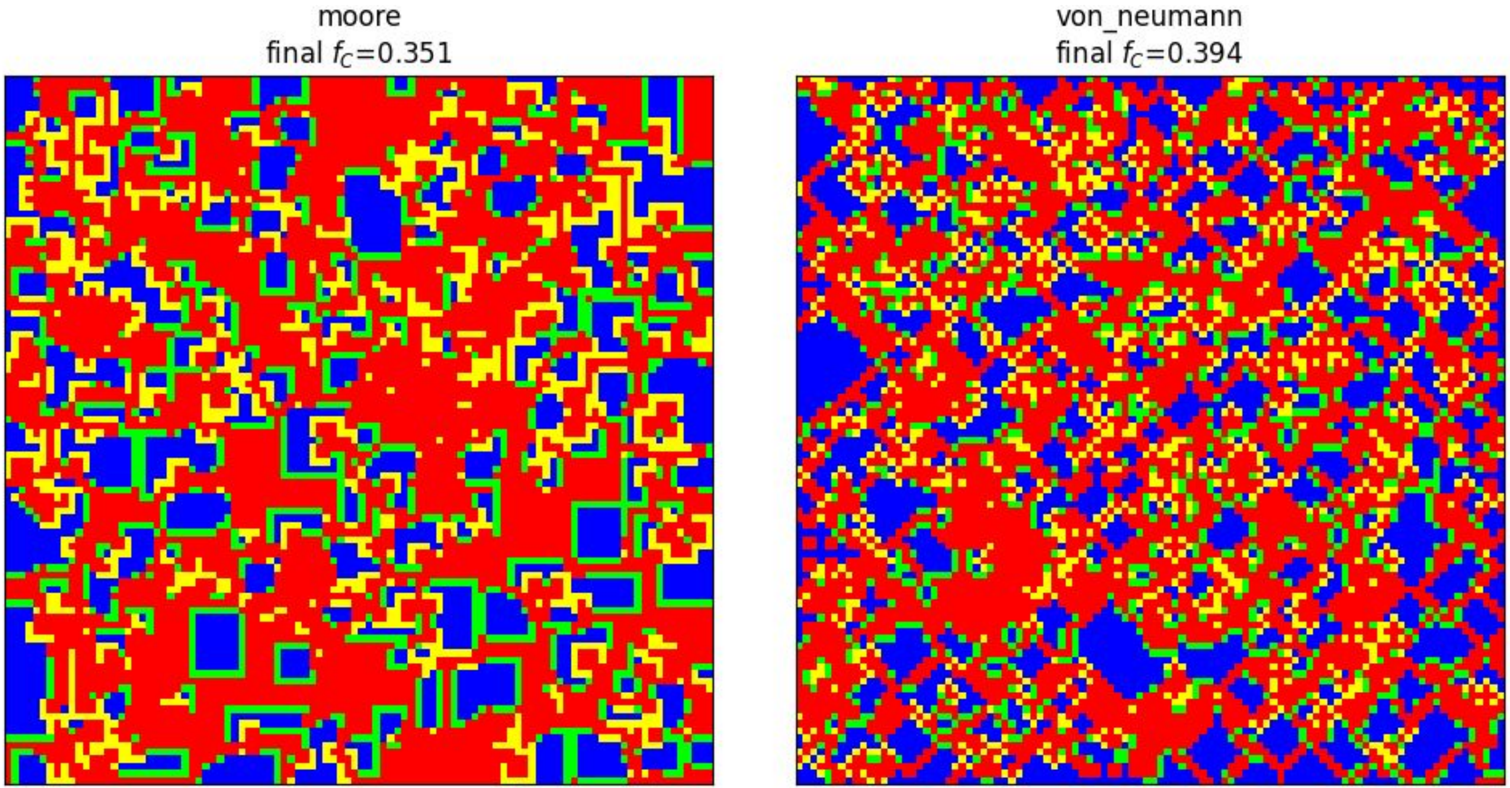
Results

- Cutoff vs periodic boundaries yield similar fraction of cooperators for b = 1.9
  - System behavior is intrinsic, i.e. not an artifact of edges
- **Practical implications:** local PD competition on an unbounded surface may look similar to bounded surface, e.g. microbes on the surface of a sphere vs a petri dish

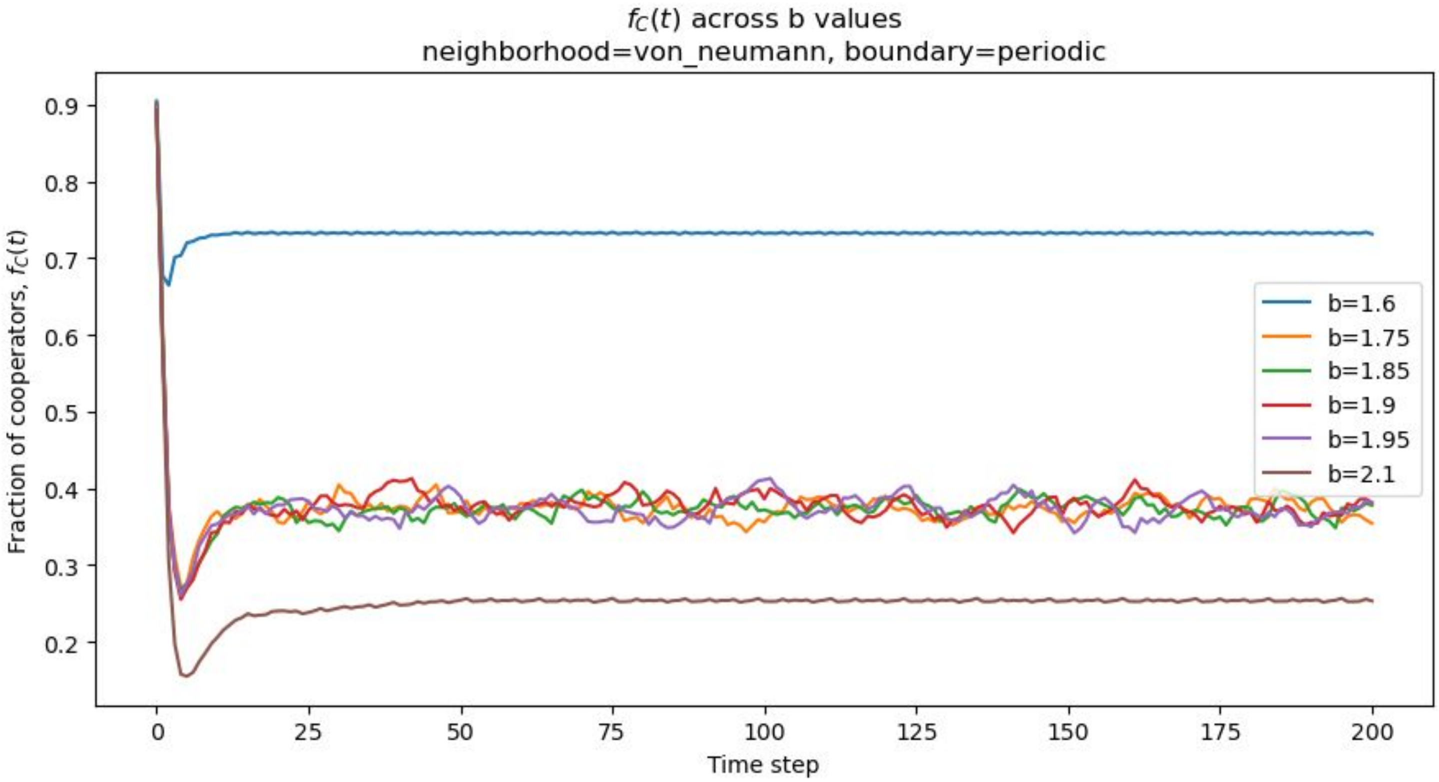
Neighborhood Variants (b = 1.9)



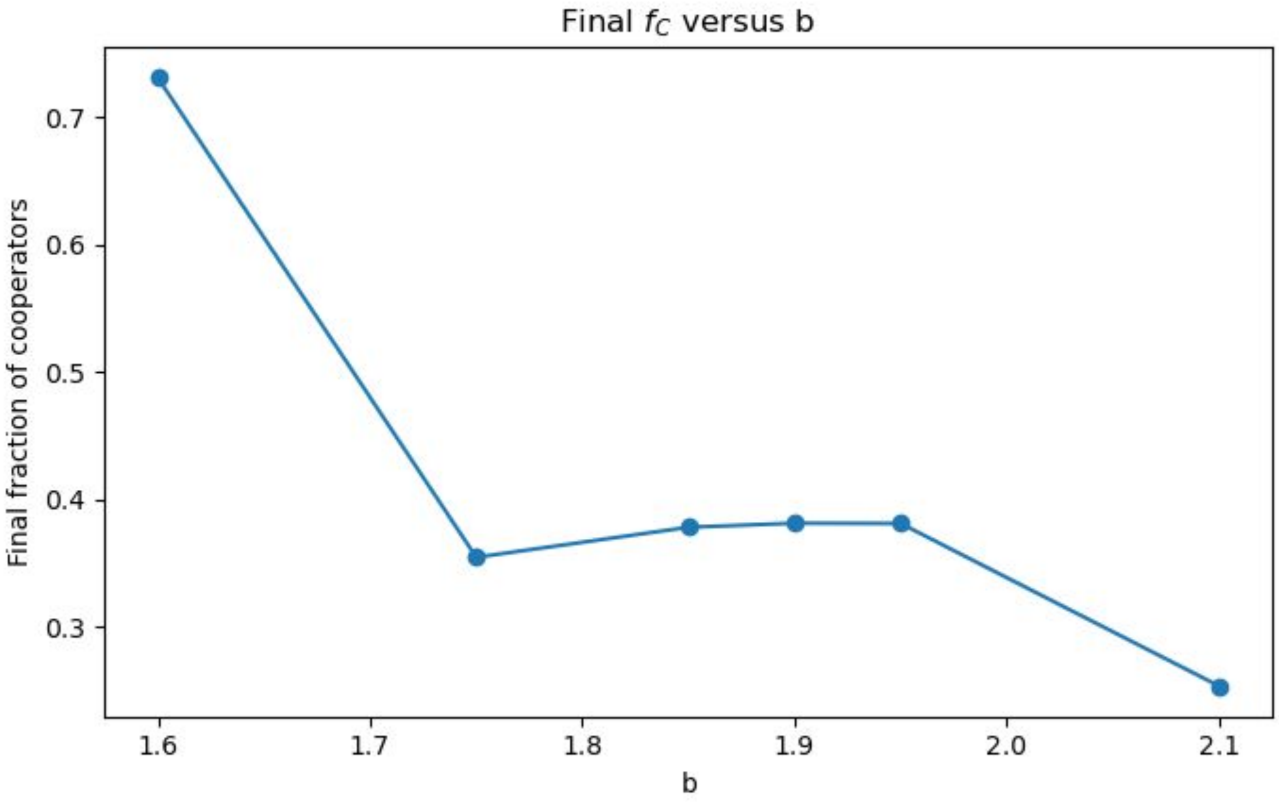
Neighborhood Variants (b = 1.9)



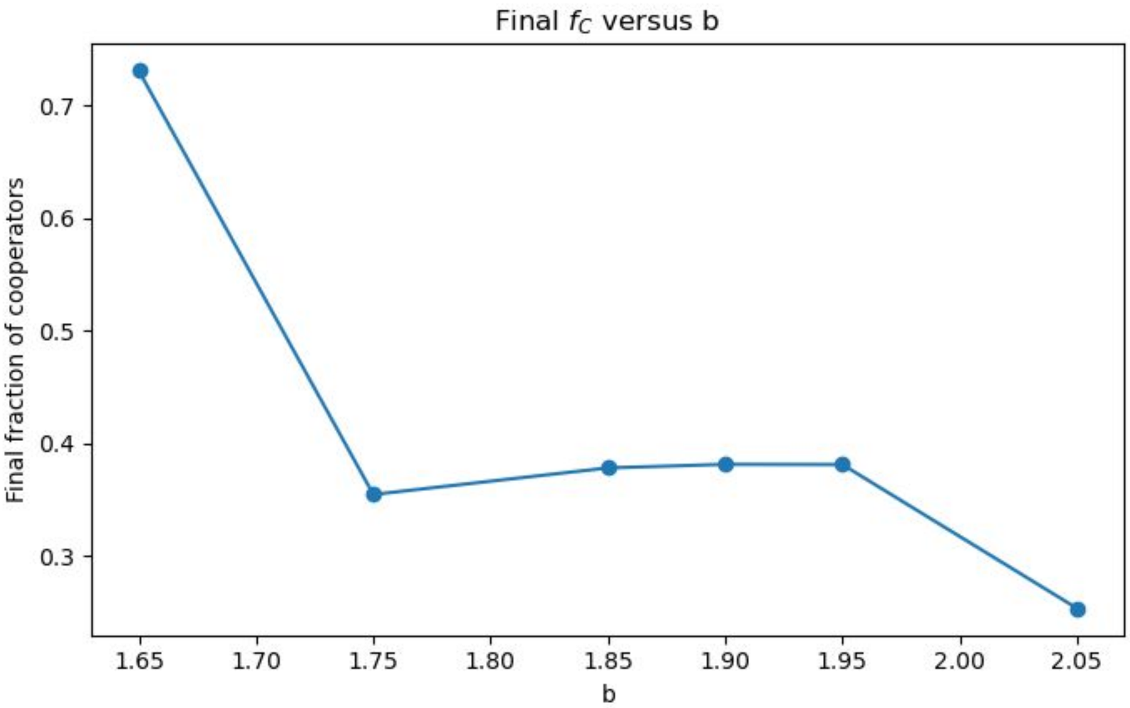
Neighborhood Variants (Vary b)



Neighborhood Variants (Vary b)



Neighborhood Variants (Vary b)



New competitive range: 1.7 < b < 2.0

Neighborhood Variants

Results

- Von Neumann neighborhood yields higher fraction of cooperators for $b = 1.9$
    - Intuition: Less “attack vectors” into communities of cooperators (4 vs 8)
- Neighborhood impacts competitive range for b
    - Moore: $1.8 < b < 2$
    - Von Neumann: $\sim 1.7 < b < 2$

Practical implications: structure of interactions influences survival of cooperation

The evolution of cooperation is dependent on both the structure of the interactions and the temptation to defect

Thank you!

Results

- **b > 1.8:** a 2x2 or larger cluster of D will grow
- **b < 1.8:** large D clusters shrink

- **b < 2:** a 2x2 or larger cluster of C will grow
- **b > 2:** large C clusters will not grow

- **1.8 < b < 2:** chaotic coexistence
- Proportion of cooperators settles to ~0.32 independent of initial conditions